

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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*A thesis submitted in fulfillment of the requirements
for the degree of Doctor of Science*

in the

Institute of High Energy Physics
Chinese Academy of Sciences



*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

0.0.1 Extraction of Differential $H \rightarrow 4\ell$ Fiducial Cross Sections

Kinematic Observables

The differential cross section measurements are performed as a function of the kinematic properties of the four-lepton system and the jet multiplicity, and are unfolded to the particle level. In this document, regarding the kinematic properties of the four-lepton system, the results are reported for the transverse momentum and rapidity of the four-lepton system ($p_T(H)$ or $p_T(4\ell)$, $|y(H)|$ or $|y(4\ell)|$), $N(\text{jets})$, and $p_T(\text{jet})$ which probe the production mechanism. The binning boundaries for each kinematic observable have been optimized in order to have a similar expected measured cross-section for every bin.

Background Model

In order to perform the differential measurements, the background is to be subtracted at the reconstruction level for each bin of the considered observable. In general, the background mass spectrum shape is not the same in all bins of the considered observable because the background does not have a resonance structure. The mass spectrum also can have non-negligible correlation with the observables.

In case of the irreducible backgrounds, the four-lepton mass spectra are modeled as the $m_{4\ell}$ template shapes using simulated events for each bin of the observable considered for the measurement. Similarly, in case of the reducible backgrounds the templates are built from the control regions in the data. This procedure is similar to the one described and used in the Ref. [Chatrchyan:2013mx].

Fractions of background events in each bin of $p_T(4\ell)$ are presented in Table 1.

TABLE 1: Estimated fraction of background events in each bin of $p_T(4\ell)$, for each final state using 2018 samples.

Background	final state	0 – 15	15 – 30	30 – 45	45 – 80	80 – 120	120 – 200	200 – 13000
qqZZ	$2e2\mu$	0.483668	0.239196	0.130151	0.105276	0.0266332	0.011809	0.00326633
qqZZ	$4e$	0.459665	0.234399	0.136986	0.114916	0.0388128	0.00989346	0.00532725
qqZZ	4μ	0.492632	0.241203	0.110977	0.109173	0.0297744	0.0129323	0.00330827
ggZZ	$2e2\mu$	0.310446	0.309859	0.191901	0.168134	0.0187793	0.000880282	0.0
ggZZ	$4e$	0.3085	0.306332	0.19247	0.17433	0.0169994	0.00136908	0.0
ggZZ	4μ	0.321332	0.319311	0.194863	0.150987	0.0128703	0.000638196	0.0
Z+X (CR)	$4l$	0.0992521	0.211153	0.209307	0.305327	0.120303	0.0483797	0.00627828

Signal Model and Statistical Procedure

In order to measure the differential cross sections within the fiducial phase space, we extract the cross sections in the bins of kinematic observables at the fiducial level following the procedure outlined in Section ???. To extract the signal yields, the signal mass spectrum is built in each cell of the response matrix using a double-sided Crystal Ball (CB) function, as was done in Ref. [Chatrchyan:2013mxa]. In practice, the same parameters of the CB function are used in each cell and the systematic uncertainties assigned on the energy scale and resolution are general enough to cover the discrepancy of the signal mass spectrum among each cell.

Examples of the fitted signal line shapes for differential bins of $p_T(H)$ from $gg \rightarrow H$ (POWHEG+JHUGEN 125 GeV 2018 sample) in the individual final states are shown in Figure 1, and examples of the efficiency matrices for $p_T(H)$ for the $2e2\mu$ final state are shown in Fig 2. Many more examples can be seen in Appendix ??.

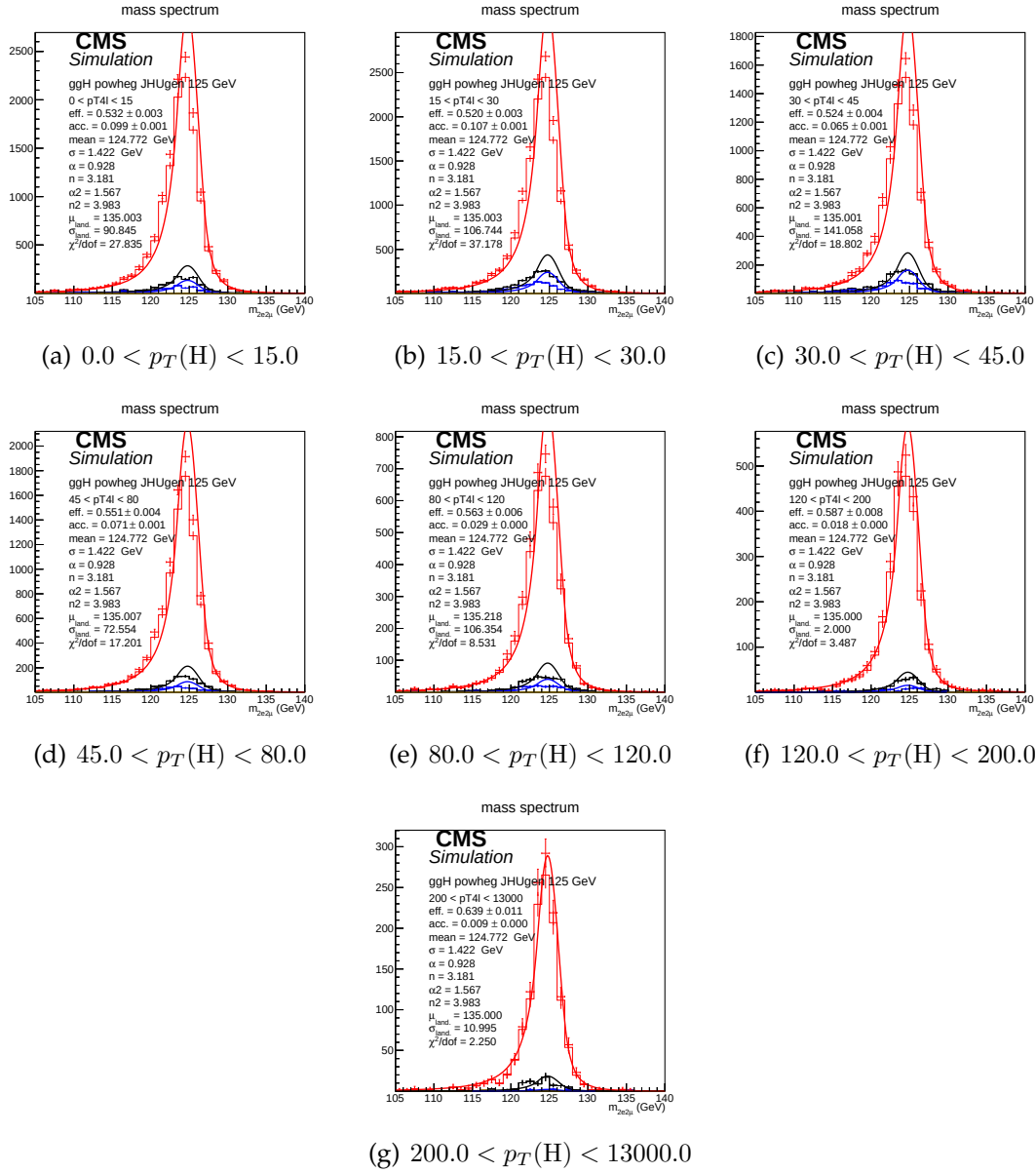


FIGURE 1: Distributions of $m(4\ell)$ in $2e2\mu$ final state for the $gg \rightarrow H$ production mode from POWHEG+JHUGEN in different bins of $p_T(H)$.

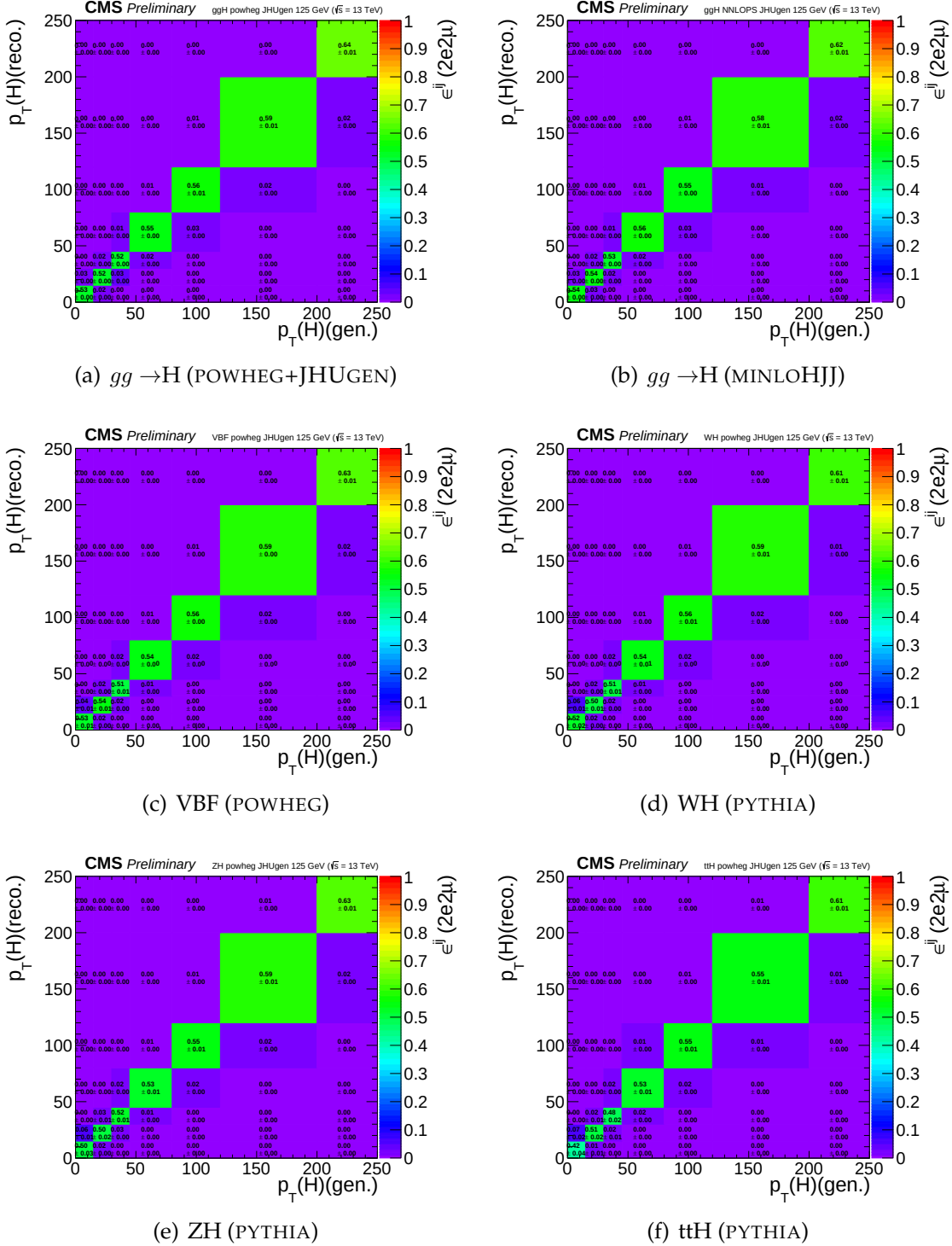


FIGURE 2: Efficiency matrices for $p_T(H)$ for different SM production modes in the $2e2\mu$ final state using 2018 samples.

Figure 3 shows the four lepton mass distributions for an Asimov dataset generated using SM cross section values and efficiencies from the $gg \rightarrow H$ production mode from POWHEG+JHUGEN, and resulting fitted values of the signal and background pdfs in different bins of $p_T(H)$ (all final states combined).

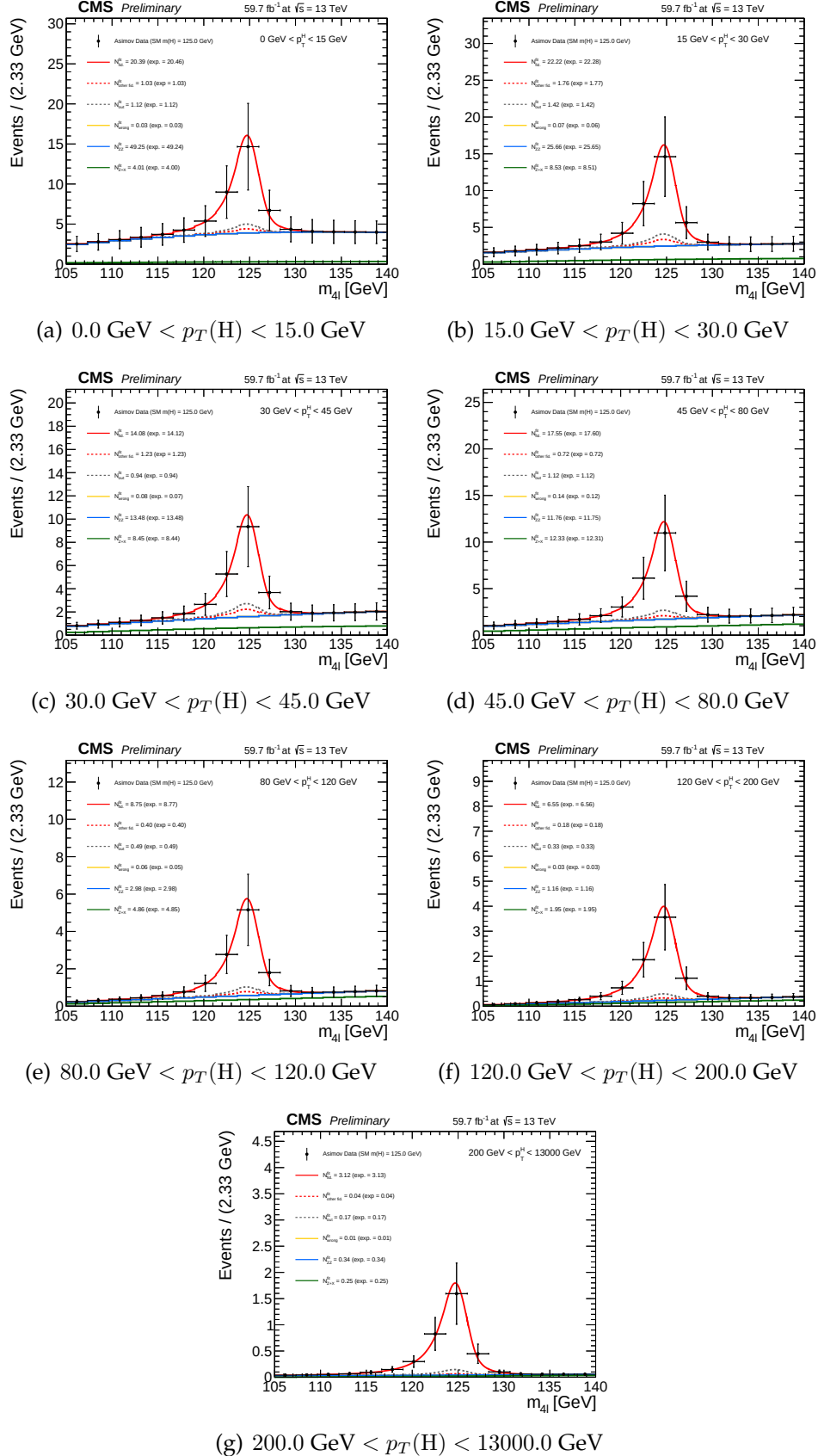


FIGURE 3: Four lepton mass distributions for an Asimov dataset generated using SM cross section values and SM efficiencies where the $gg \rightarrow H$ prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different bins of $p_T(H)$ (all final states combined).

0.1 Systematic Uncertainties

0.1.1 Irreducible ZZ Background Uncertainties

The $qq \rightarrow ZZ$ and $gg \rightarrow ZZ$ four lepton mass spectrum and yields are taken from MC samples. The estimation from the MC samples is used as the initial value for the fit, for which we use a model derived from a NLO generation of $qq \rightarrow ZZ \rightarrow 4\ell$ using POWHEG generator. For $gg \rightarrow ZZ$ instead, we use the prediction from MCFM 6.7 generator. The mass spectrum used in the fits are fixed to the MC estimate, while the yields are left varying within the uncertainty.

PDF+ α_s and QCD uncertainties for $pp \rightarrow ZZ \rightarrow 4\ell$ at NLO and $gg \rightarrow ZZ \rightarrow 4\ell$ are evaluated using MCFM. We use the $2e2\mu$ final state and the following fiducial cuts for leptons: $m_{ee} > 12$, $m_{\mu\mu} > 12$, electrons $p_T > 7$ and $|\eta| < 2.5$, and muons $p_T > 5$ and $|\eta| < 2.4$. The minimal jet-lepton and lepton-lepton ΔR_{min} -distance were relaxed, i.e. set to zero. The cross sections are calculated inclusively in the number of jets found at NLO. The PDF+ α_s and QCD scale uncertainties are treated as uncorrelated. In case of the differential measurements, these are computed for each bin of each of the considered observables separately.

For the estimation of the PDF+ α_s systematic errors, we use the PDF4LHC prescription. The three PDF sets used are CT10, MSTW08, NNPDF. The obtained results are summarized in Figures 4.

For estimation of QCD scale systematic errors, we calculate variations in the differential cross section $d\sigma/dm_{4\ell}$ as we change the renormalization and factorization scales by a factor of two up and down from their default setting $\mu_R = \mu_F = m_Z$. The obtained results are summarized in Figure 5.

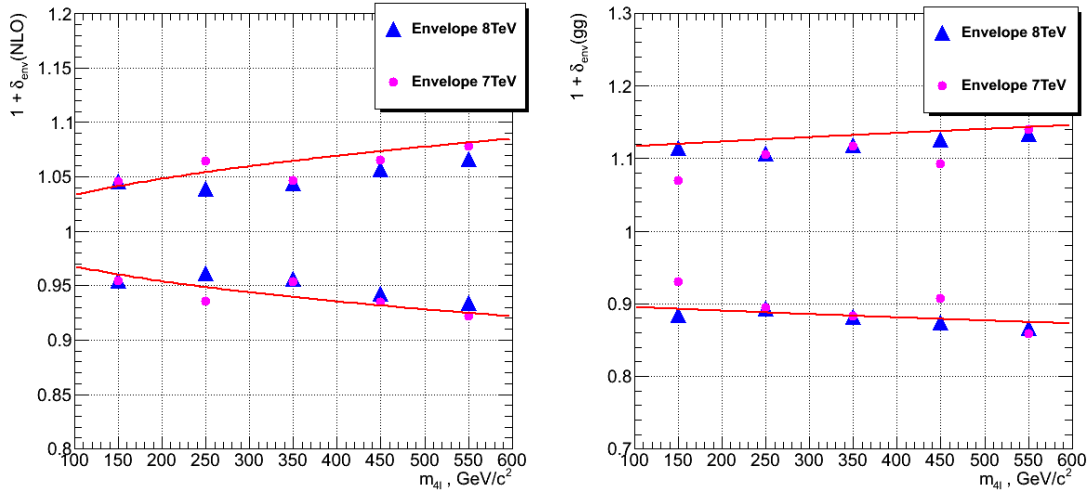


FIGURE 4: PDF+ α_s uncertainties for (left) $pp \rightarrow ZZ \rightarrow 4\ell$ at NLO and (right) $gg \rightarrow ZZ \rightarrow 4\ell$ processes. The points are evaluated uncertainties. The curves are the fit systematic error $\kappa(m_{4\ell})$ to be used in the statistical analysis. These errors are driven by two independent nuisance parameters pdf_qqbar for $pp \rightarrow ZZ \rightarrow 4\ell$ at NLO and pdf_gg for $gg \rightarrow ZZ \rightarrow 4\ell$.

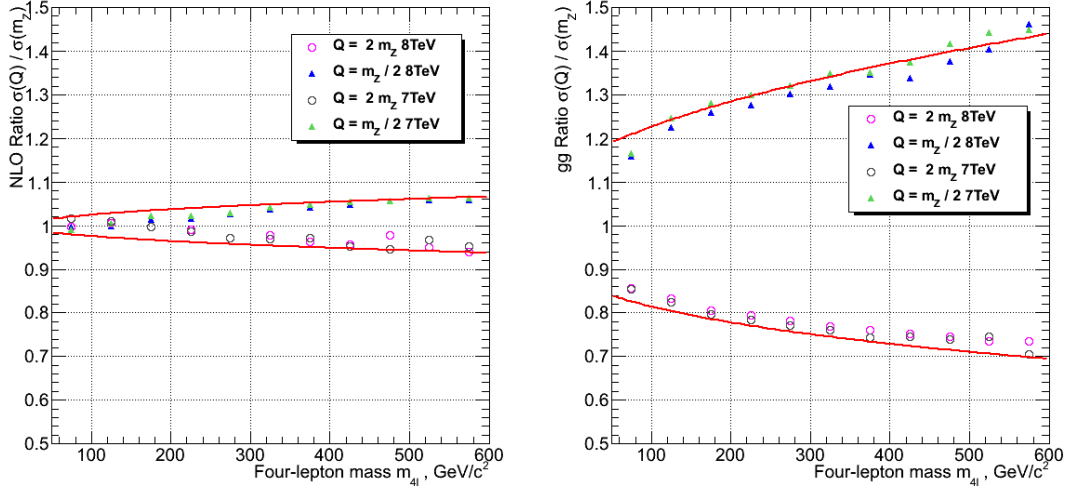


FIGURE 5: QCD scale uncertainties for (left) $pp \rightarrow ZZ \rightarrow 4\ell$ at NLO and (right) $gg \rightarrow ZZ \rightarrow 4\ell$ processes. The points are evaluated uncertainties. The curves are the fit systematic error $\kappa(m_{4\ell})$ to be used in the statistical analysis. These errors are driven by two independent nuisance parameters $QCDscale_VV$ for $pp \rightarrow ZZ \rightarrow 4\ell$ at NLO and $QCDscale_ggVV$ for $gg \rightarrow ZZ \rightarrow 4\ell$.

0.1.2 Uncertainties on the Reducible Background Predictions

For the determination of the shape of the $m_{4\ell}$ distribution for the channels where $Z_2 \rightarrow ee$, the two channels $4e$ and $2\mu 2e$ are treated together. The uncertainty of the $m_{4\ell}$ shape is driven by that of the 3P1F component, of which the statistics is limited. To simplify the statistical analysis, the uncertainty of the $m_{4\ell}$ shape is included a global normalization uncertainty of 20%.

For the determination of the shape of the $m_{4\ell}$ distribution for the channels where $Z_2 \rightarrow \mu\mu$, the two channels 4μ and $2e 2\mu$ are treated together. The shape is rather well constrained, and any uncertainty on the $m_{4\ell}$ distribution is absorbed in the large systematic uncertainty of 40% set on the event yields.

The channels $2e 2\mu$ (where the Z_1 is made from two electrons) and $2\mu 2e$ (where the Z_1 is made from two muons) are treated together. The mass shape used for this channel is the average of the two shapes presented above, weighted by the event yields expected in the two channels, and a global normalization uncertainty of 25% is used.

0.1.3 Uncertainties on Signal Shape and Yields

The uncertainties affecting the shape of the signal arise from the non perfect knowledge of the lepton scale and resolution, which originates from the dependency of the scale and resolution on p_T , η and from run conditions. For muons, the energy scale uncertainty is less than 0.1%, which propagates to about 0.1% on the 4ℓ mass for 4μ events. For electrons, the systematics due to the energy scale dependency

on p_T are 0.3% systematic for $4e$ final state and extra 0.2% systematic for $2e2\mu$ final state on the mean of the double-sided Crystal Ball function. In addition, the uncertainty on the resolution is estimated to be 20%.

For both integrated and differential cross sections measurements, the uncertainty on the signal yields comes from uncertainties on the luminosity and imperfect knowledge on single lepton efficiency which are taken from [Chatrchyan:2013mxq, Aad:2013xqa]. For the "wrong" signal process, its mass spectrum is taken from $WH, ZH, t\bar{t}H$ MC samples and parameterized using a Landau function. The mean and sigma of the Landau function are allowed to float to cover the difference in the shape between $WH, ZH, t\bar{t}H$ processes. The expected normalization on the "wrong" signal process is taken from SM prediction, i.e., the sum of contributions from $WH, ZH, t\bar{t}H$ processes. We allow this normalization varies according to a log-uniform distribution, of which the $1-\sigma$ variation corresponds to enhance or suppress the expected normalization by one order of magnitude.

0.1.4 Uncertainties on Jet Energy Scale

For differential cross section measurement as a function jet multiplicity, the uncertainty due to jet energy scale has non-negligible effect which can effect the number of reco jet distribution while has no impact on number of gen jet distribution. Figure ?? shows the number of reco-jet distribution variation when the reco jet energy is scaled up by $1-\sigma$. As a result, the efficiency of events with number $N_{gen-jet}$ gen jets being reconstructed as events with $N_{reco-jet}$ reco jets will vary according to the variation of the jet energy scale. To correlate the variation due to reco-jet energy scale for signal and background events in different reco bin, we introduce a nuisance parameter θ_{JES} with unit Gaussian distribution,

$$\text{eff}(N_{reco-jet}, N_{gen-jet}|\theta_{JES}) = (1 + \lambda(N_{reco-jet}, N_{gen-jet})\theta_{JES}) \times \text{eff}(N_{reco-jet}, N_{gen-jet}|\theta_{JES} = 0). \quad (1)$$

If define $\theta_{JES} = \pm 1$ corresponding to $1-\sigma$ variation of the reco-jet energy scale, then the parameter $\lambda(N_{reco-jet})$ can be calculated using the nominal efficiency and efficiency at $1-\sigma$ variation of the reco-jet energy scale as

$$\begin{aligned} & \lambda(N_{reco-jet}, N_{gen-jet}) \\ = & \frac{\text{eff}(N_{reco-jet}, N_{gen-jet}|\theta_{JES} = \pm 1) - \text{eff}(N_{reco-jet}, N_{gen-jet}|\theta_{JES} = 0)}{\text{eff}(N_{reco-jet}, N_{gen-jet}|\theta_{JES} = 0)} \end{aligned} \quad (2)$$

In practice, $\lambda(N_{reco-jet}, N_{gen-jet})$ is calculated using MC samples or control region data for each reco bin for both signal and background processes. Examples of $\lambda(N_{reco-jet}, N_{gen-jet})$ factors can be seen in Table 2.

TABLE 2: Percent change in events when increasing the jet energy scale by 1σ for various signal model (all final states combined).

Sample	N(jets)=0	N(jets)=1	N(jets)=2	N(jets) $\geq$ 3
gg $\rightarrow$ H (POWHEG+JHUGEN) 125 GeV	-0.027	0.024	0.075	0.104
gg $\rightarrow$ H (MINLOHJJ) 125 GeV	-0.023	0.028	0.073	0.120
VBF (POWHEG) 125 GeV	-0.088	-0.028	0.025	0.081
WH (PYTHIA) 125 GeV	-0.050	-0.028	0.029	0.097
ZH (PYTHIA) 125 GeV	-0.032	-0.034	0.032	0.071
ttH (PYTHIA)125 GeV	0.000	-0.157	-0.053	0.010

0.1.5 Uncertainties on the Pileup Jet ID Efficiency and the Jet Energy Resolution

The uncertainties on the jet ID efficiency also contribute to the uncertainties on the jet counting (i.e., applying or not applying the jet ID cut should change the unfolding matrix). The Pileup Jet ID efficiency measured in data agrees with simulation very well in all regions except for $|\eta| > 3.0$, as studied in Ref. [Aad:pile_up_jet_ID]. Following those Pileup Jet ID efficiency studies, we have applied the correction factors for jets with $|\eta| > 3.0$ in three bins of p_T : factor 0.90 for $30 < p_T < 36$, factor 0.94 for $36 < p_T < 50$, and factor 0.97 for $p_T > 50$. The remaining uncertainty on the jet ID efficiency is then small compared to the JES uncertainty and can be neglected.

In addition, we have also checked if the uncertainties on the jet energy resolution (JER) contribute to the uncertainties on the jet multiplicity measurement. We have estimated the JER uncertainties according to Jet/MET recommendation, and we have found that the uncertainty due to JER is much smaller than (and covered by) the JES uncertainty. For this reason, the JER uncertainty is not added separately. Comparisons of the JER uncertainty and JES uncertainty for different production modes is shown in Figure ?? in Appendix ??.

0.1.6 Uncertainty Due to Model Dependence

One of the major goals of this analysis is to provide the model-independent measurements of $H \rightarrow 4\ell$ integrated and differential cross sections. To achieve a high degree of the model independence, we introduced isolation in the definition of the fiducial phase space and excluded the "wrong" signal contributions from the fiducial phase space the definition. However, there is unavoidable remaining model dependence both in the integrated and differential cross section measurements. The efficiency for integrated cross section measurement is consistent between different models with about 10% uncertainty, and this uncertainty can be up to 25% in case of the number of jet differential cross section measurements. In addition, the non-fiducial signal also contributes to model dependence. A graphical representation of the model dependence of this ratio is shows in the Figure 6.

To properly include the correlations of all the sources of the model dependence, we extract the systematic uncertainty due to signal model dependence in the following way: for each signal model, we use the corresponding MC sample to extract the fiducial signal efficiencies, non-fiducial signal fraction, and $\lambda(N_{reco-jet}, N_{gen-jet})$ in case of the number of jets differential measurement. The integrated and differential cross section results are extracted with these inputs generated according to the SM composition of the Higgs boson production modes. We then extract the results using the inputs of individual models which correspond to each of the Higgs production mechanisms in proton-proton collisions, rare $H \rightarrow 4\ell$ decay through virtual photon(s) and exotic spin/parity models. The largest difference between the results of these individual measurements and the result corresponding to the SM-like prediction we quote as the systematic uncertainty associated with the model dependence.

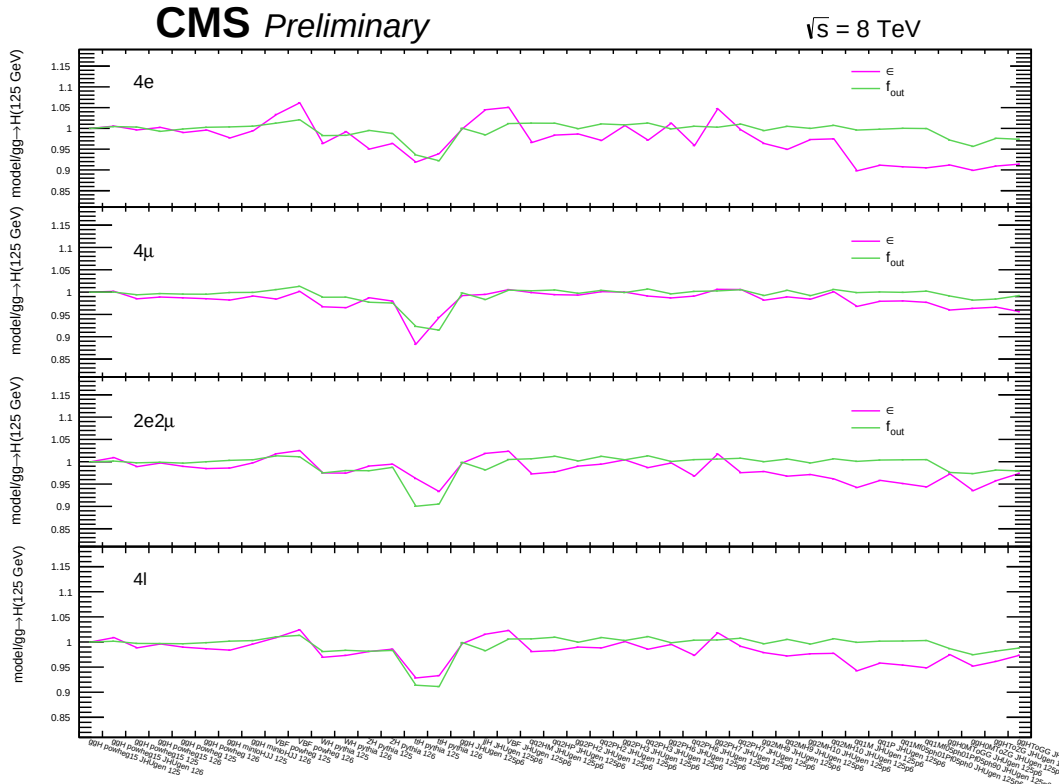


FIGURE 6: Ratio of integrated ϵ and f_{nonfid} for all signal models to the integrated ϵ and f_{nonfid} from the $gg \rightarrow H$ production mode (POWHEG+JHUGEN 125 GeV).

Additionally, we assess the model dependence uncertainty by taking into account the experimental constraints from the existing CMS measurements. We consider the Higgs boson to be a SM-like spin-zero boson with SM-like production mechanisms (all exotic decays and spin-parity models excluded), and we allow the cross-section for the individual production mechanisms to vary within ranges that

TABLE 3: Summary of all systematic uncertainties affecting $H \rightarrow 4\ell$ cross section measurements.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.2% (7 TeV), 2.6% (8 TeV)
Lepton identification/reconstruction efficiencies	4–10%
Background related uncertainties	
QCD scale ($q\bar{q} \rightarrow ZZ$, $gg \rightarrow ZZ$)	3–24%
PDF set ($q\bar{q} \rightarrow ZZ$, $gg \rightarrow ZZ$)	3–7%
Reducible background (Z + X)	20–40%
Jet resolution and energy scale	2–16%
Signal related uncertainties	
Lepton energy scale	0.1–0.3%
Lepton energy resolution	20%
Jet energy scale and resolution	3–12%
Combinatorial signal-induced contribution	
Effect on the final measurement	4–11%
Model dependence	
With exp. constraints on prod. modes & exotic models	1–5%
No exp. constraints on prod. modes & exotic models	7–25%

correspond to the allowed 95% CL intervals from the latest CMS combined results on the Higgs boson properties. In this case, the systematic uncertainty associated with the model dependence is 3–5% in case of the number of jets differential measurements, and it is smaller than 1% for all the other measurements. The results can be found in Appendix ??.

As a test of the method, we also generate an alternative Asimov dataset according to a model which consists purely of ZH production and with a cross section set to the total SM cross section. We then unfold this dataset using the nominal SM values for the efficiencies and out of acceptance fractions. The results of this test are shown in Appendix ??.

The imprecision in the cross section measurements due to the m_H being fixed in the fit procedure is estimated from simulation to be 2% per 0.6 GeV shift in mass. As the uncertainty on the CMS mass measurements is 0.3 GeV, the additional systematic uncertainty on the cross section measurement would correspond of 1%, and is covered by the larger already existing systematic uncertainties. The Summary of all systematic uncertainties affecting $H \rightarrow 4\ell$ cross section measurements is tabulated in Table 3.